

check in 3

damian rojas

2026-05-28

Set-up

```
library(tidyverse)
library(ggeffects)
library(janitor)
library(ggplot2)
library(dplyr)
library(plotly)
library(viridis)
library(hrbrthemes)
library(ggbeeswarm)
#beeswarm package from failed beeswarm plot
```

Read in data from personal data project

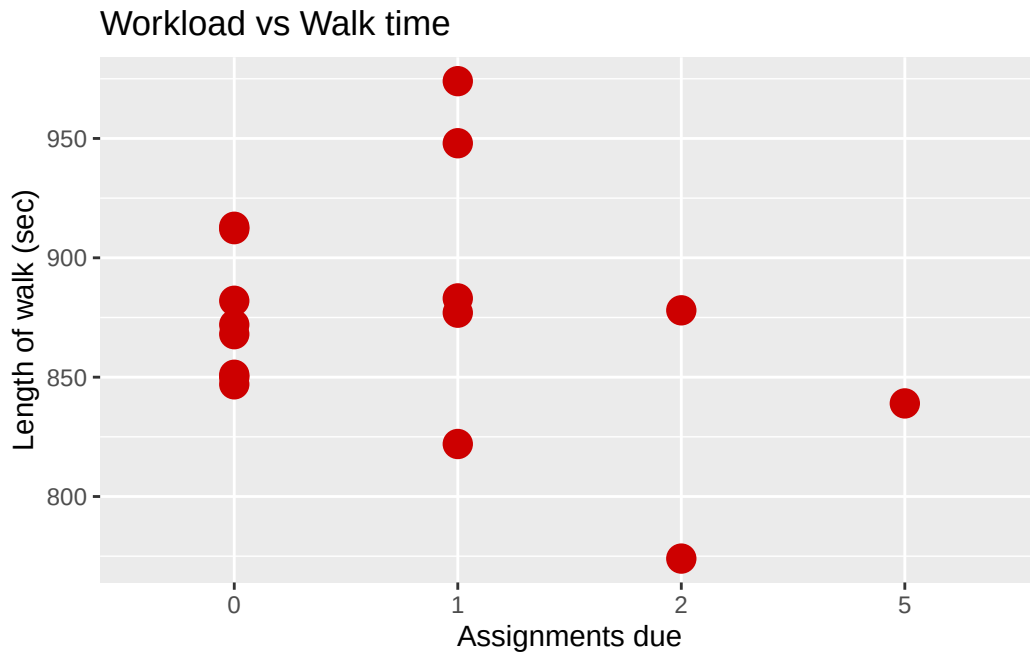
```
walks <- read.csv("walks_data.csv") |>
#from HW 2
mutate(work_due = as_factor(work_due))
      #factor so data is listed in order of appearance, cleans up repeats
#create graph, start with gg plot as base layer
# x = categorical predictor
ggplot(walks, aes(x = work_due,
                  y = walk_time_s)) +
# y = response variable (in seconds to avoid graph looking skewed)
# scatter plot to compare relationship between predictor and response variable
geom_jitter(height = 0,
            width = 0,
```

```

    shape = 16,
    size=5,
    color="red3") +

#label graph
labs(title="Workload vs Walk time",
     x = "Assignments due",
     y = "Length of walk (sec)") +
#change theme
theme_gray()

```



```

#from HW2 to show potential relationship between degrees and walk time
#collect personal data
walks |>

mutate(WEATHER_F = as_factor(WEATHER_F))

```

	DATE	WEATHER_F	TIME.Start.	WIND.SPEED	WIND.DIR.	DEADTIME.min.sec.	TIME.End.
1	4/28	66	7:06 PM	4	AWAY	0:00	7:22 PM
2	4/29	64	5:54 PM	4	AGAINST	0:46	6:09 PM
3	4/30	62	7:29 PM	8	AWAY	0:49	7:44 PM
4	5/3	60	7:23 PM	11	AGAINST	0:14	7:38 PM

5	5/4	60	7:07	8	AGAINST	0:00	7:28
6	5/6	64	8:04	6	AWAY	0:16	8:18
7	5/12	60	7:42	2	AGAINST	0:24	7:57
8	5/12	60	8:05	2	AWAY	0:00	8:22
9	5/13	65	6:37	6	AGAINST	0:15	6:52
10	5/13	63	7:16	5	AWAY	0:07	7:31
11	5/14	66	3:43	6	AWAY	0:00	3:55
12	5/15	64	1:58	7	AGAINST	0:00	2:12
13	5/19	73	6:44	8	AGAINST	0:27	6:59
14	5/19	72	7:11	6	AWAY	0:19	7:25
15	5/20	67	8:07	4	AGAINST	1:00	8:22
16	5/20	67	8:26	4	AWAY	0:00	8:40

	work_due	walk_time	walk_time_s
1	1	15:48	948
2	0	14:10	850
3	2	14:38	878
4	1	14:43	883
5	0	14:28	868
6	1	13:42	822
7	1	14:37	877
8	1	16:14	974
9	0	15:13	913
10	0	15:12	912
11	2	12:54	774
12	5	13:59	839
13	0	14:32	872
14	0	14:11	851
15	0	14:07	847
16	0	14:42	882

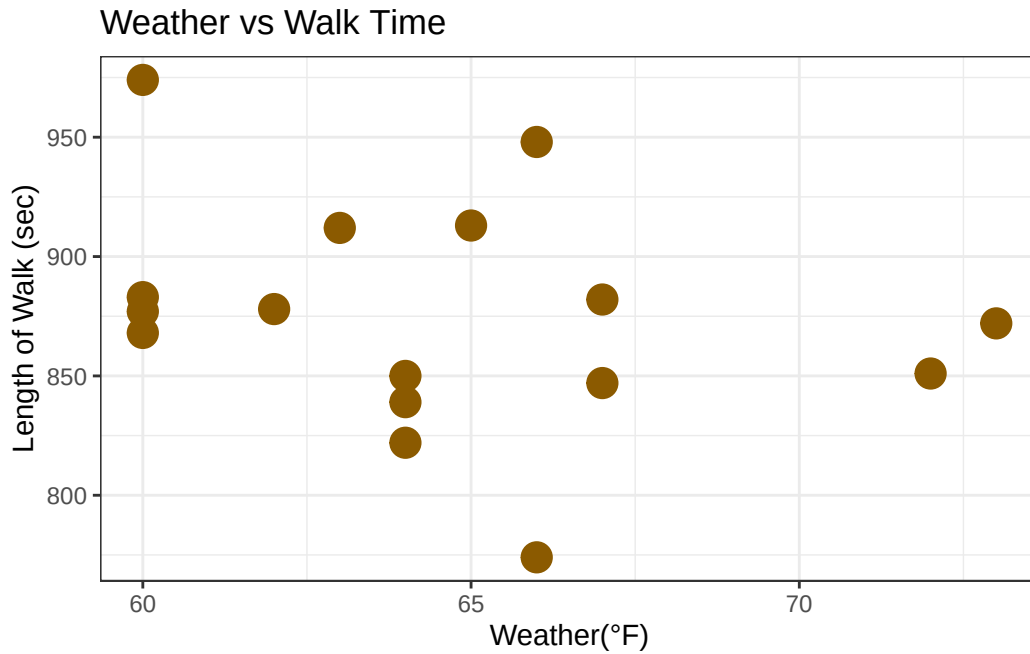
```
#create graph, start with gg plot as base layer
#x=continuous predictor, weather
#y= length of walk
#take degrees F of datqa sheet, abel in axis
ggplot(walks, aes(x = WEATHER_F,
                  y = walk_time_s)) +
#scatter plot to compare relationship between continuous predictor and response
geom_point(color="orange4", size = 5) +

#label graph
labs(title = "Weather vs Walk Time",
      x = "Weather(°F)",
```

```

y = "Length of Walk (sec)" +
#change theme
theme_bw()

```



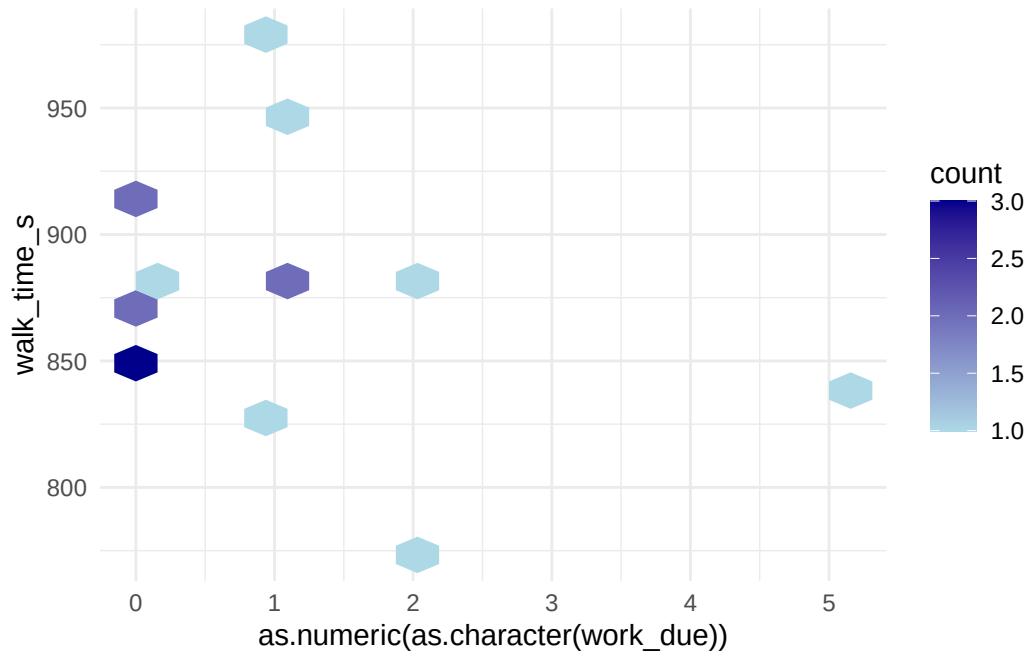
Attempting new visualization (hexbin)

```

#hexbin plot

#base ggplot, dataset = walks
#x stored as numeric factor form work due
#y = walk time in seconds
ggplot(walks, aes(x = as.numeric(as.character (work_due)), y = walk_time_s)) +
  #hexbin geometry frmo ggplot2
  #bins = 16 for gut-check
  geom_hex(bins = 16) +
  #set gradient fill from light blue (low) to dark blue for high values
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  theme_minimal()

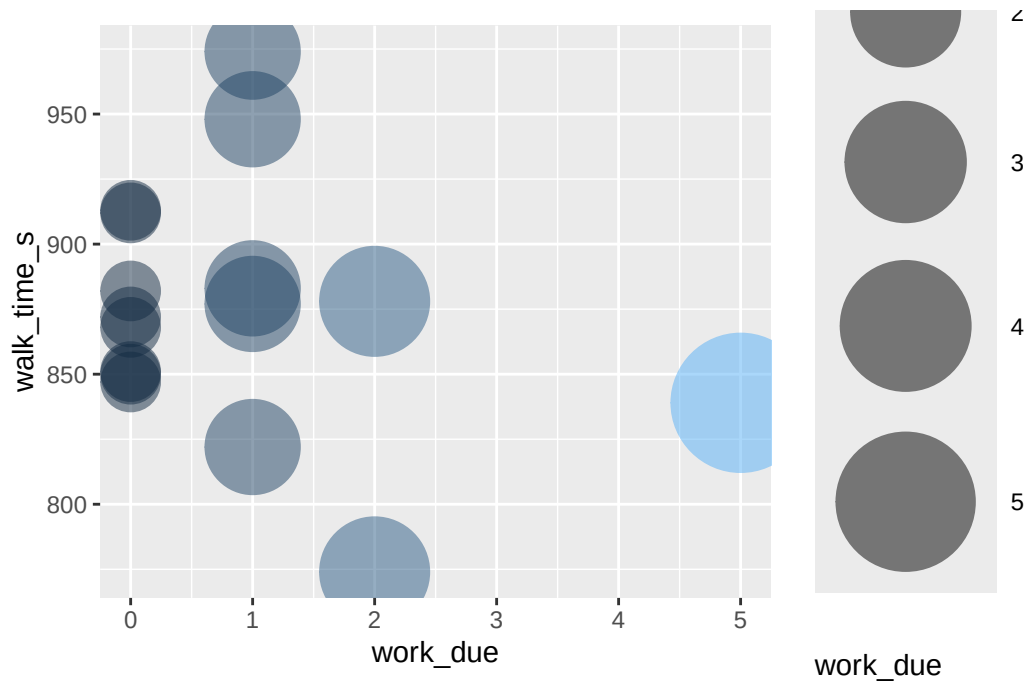
```



```
#doesn't look great try something else
```

Attempting new visualization (bubblechart)

```
#walks data
walks |>
  #arrange data by descending order for work due
  arrange(desc(work_due)) |>
  #store work due as numeric value instead of factor
  mutate(work_due = as.numeric(as.character(work_due))) |>
  #base ggplot, x = work due, y = walk time in seconds
  ggplot(aes(x=work_due, y = walk_time_s, size = work_due, color = work_due)) +
  #geom point for bubblechart shape
  geom_point(alpha=0.5) +
  #scale by work due,
  scale_size(range = c(10,24), name = "Work Due")
```



Attempting new visualization

(violin plot)

```
walks |>
  #mutate work_due as a factor
  mutate(work_due = factor(work_due)) |>
  #base gg plot
  ggplot(aes(
    # x = work due
    x = work_due,
    #y is walk time in seconds
    y = walk_time_s,
    #color by work due
    color = work_due
  )) +
  #violin plot geometry
  geom_violin(
    #thickness of lines
    linewidth = 1,
    #drop equal false to not drop data sets with fewer than 2 values
    drop = FALSE ) +
```

```
#set color by work due
scale_color_viridis_d("work_due") +
theme_minimal()
```

Warning: Cannot compute density for groups with fewer than two datapoints.

